

National University of Computer and Emerging Sciences, Lahore Campus



Course: Linear Algebra
 Program: BS (SE)
 Duration: 60 Minutes
 Paper Date: 29-02-2024
 Section: ALL
 Exam: Sessional-1

Course Code: MT1004
 Semester: Spring 2024
 Total Marks: 40
 Weight: 15%
 Page(s): 1
 Roll No:

Instruction/Notes:

1. Programmable calculators are not allowed.
2. Do all the questions in the given order as mentioned in the paper.
3. Do not ask any questions regarding contents. If you feel any essential information missing make your own assumption.

Question # 1 (CLO-1) [10] The given matrix represents an augmented matrix for a linear system.

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -4 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}$$

- a. Write the corresponding set of linear equations for the system.
- b. Use Gaussian elimination to solve the linear system.

Question # 2 (CLO-1) [10] Given the matrix $\begin{bmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{bmatrix}$

- a. Find the minor M_{21} and cofactor C_{21} .
- b. Evaluate the determinant by cofactor expansion.

Question # 3 (CLO-1) [10]

- a. Find an equation of plane passing through a point $P(1, 1, 4)$ having normal vector $\vec{n} = (1, 9, 8)$.
- b. Also find the distance between the point $Q(-1, -1, 2)$ and the plane. Using the equation of the plane in part (a).

Question # 4 (CLO-1) [10] Find the vector component of $\vec{u}$ along $\vec{a}$ and the vector component of $\vec{u}$ orthogonal to $\vec{a}$ given that $\vec{u} = (2, 1, 1, 2)$ & $\vec{a} = (4, -4, 2, -2)$.

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 Total Marks: 40
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 Page(s): 1
 Roll No: 221-7971

Instruction/Notes:

1. Programmable calculators are not allowed.
2. Do all the questions in the given order as mentioned in the paper.
3. Do not ask any questions regarding contents. If you feel any essential information missing make your own assumption.

Question # 1 (CLO-1) [10] The given matrix represents an augmented matrix for a linear system.

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 2 & 7 & 8 \\ 0 & 4 & -2 & 5 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\left[\begin{array}{cccc} 1 & 3 & 0 & 2 \\ -2 & -4 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{array} \right]$$

$$\begin{aligned} x_1 + 3x_2 + 0x_3 &= 2 \\ -2x_1 - 4x_2 + 7x_3 &= 4 \\ 3x_1 + 5x_2 + 2x_3 &= 1 \\ x_1 - x_2 + 2x_3 &= -3 \end{aligned}$$

- a. Write the corresponding set of linear equations for the system.
- b. Use Gaussian elimination to solve the linear system.

$$\begin{aligned} x_3 &= 11/16 \\ x_2 &= 5/32 \\ x_1 &= -84/32 \end{aligned}$$

Question # 2 (CLO-1) [10] Given the matrix $\begin{bmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{bmatrix}$

- a. Find the minor M_{21} and cofactor C_{21} .
- b. Evaluate the determinant by cofactor expansion.

Question # 3 (CLO-1) [10]

- a. Find an equation of plane passing through a point $P(1, 1, 4)$ having normal vector $\vec{n} = (1, 9, 8)$.
- b. Also find the distance between the point $Q(-1, -1, 2)$ and the plane. Using the equation of the plane in part (a).

$$x + 9y + 8z = 42$$

$$36/\sqrt{146}$$

Question # 4 (CLO-1) [10] Find the vector component of $\vec{u}$ along $\vec{a}$ and the vector component of $\vec{u}$ orthogonal to $\vec{a}$ given that $\vec{u} = (2, 1, 1, 2)$ & $\vec{a} = (4, -4, 2, -2)$.

$$\begin{aligned} \text{Proj} &= \left(\frac{1}{5}, -\frac{1}{5}, \frac{1}{10}, -\frac{1}{10} \right) \\ \text{Orthogonal} &= \left(\frac{9}{5}, \frac{6}{5}, \frac{9}{10}, \frac{21}{10} \right) \end{aligned}$$

Linear Algebra (MT1004)

Sessional-I Exam

Date: September 21st 2024

Course Instructor(s)

Dr. Akhlaq Ahmad

Dr. Tayyaba Naz

Dr. Nazish Iftikhar

Dr. Sonia Hanif

Dr. Komal Hassan

Mr. Muhammad Rizwan

Ms. Maria Shabir

Total Time (Hrs): 1 Hour

Total Marks: 40

Total Questions: 1

Roll No

Section

Student Signature

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Attempt all the questions.

CLO #1: Use concept of elementary row operations to find the inverse of square matrices, determinant of a matrix and solving the system of linear equations.

Q#1. (a): Consider the following system in unknowns x, y and z . For what value(s) of "a, b" does the following system have no solution? Unique solution? Infinitely many solutions? [10 marks]

no: $a = \pm 3$ and $b \neq \pm 3$ or $b \neq a$
 uni: $a \neq \pm 3$
 inf: $a = \pm 3$ and $b = \pm 3$ or $b = a$.

$$\begin{aligned} x - 2y &= 1 \\ x - y + az &= 2 \\ ay + 9z &= b \end{aligned}$$

$R_2 - 3R_1, R_2 (-1/7),$
 $R_1 - 4R_2$

b): Use **Inversion Algorithm** to find the inverse of $A = \begin{bmatrix} 1 & 4 \\ 3 & 5 \end{bmatrix}$. Also write down A^{-1} as a product of elementary matrices $A^{-1} = E_k E_{k-1} \dots E_3 E_2 E_1$. [10 marks]

$\begin{bmatrix} -5/7 & 4/7 \\ 3/7 & -1/7 \end{bmatrix}$

c): Evaluate the given determinant by using row reduction method. [5 marks]

check

$$\begin{vmatrix} 2 & -1 & 5 & 5 \\ 3 & 1 & 2 & 4 \\ -1 & -3 & 8 & 0 \\ 1 & 1 & 2 & -1 \end{vmatrix} = 14$$

d): Find the values of α for which $\det A = 0$. [3 marks]

$\alpha = -2, \alpha = 4, \alpha = 3$

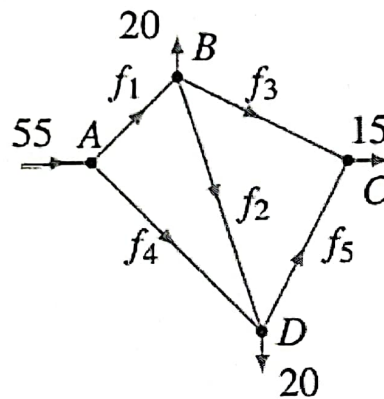
$$|A| = \begin{vmatrix} \alpha - 4 & 0 & 0 \\ 0 & \alpha & 2 \\ 0 & 3 & \alpha - 1 \end{vmatrix} = \alpha^3 - \frac{3}{\alpha} \alpha^2$$

e): Identify the row operation corresponding to E and verify that the product EM results from applying the row operation to M , where E is an elementary matrix. [2 marks]

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$M = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix} \checkmark$$

f): A proposed network of irrigation canals is described in the accompanying diagram. At peak demand, the flows at interchanges A, B, C, and D are as shown. [10 marks]



Set up a linear system whose solution provides the unknown flow rates and also solve the system for the unknown flow rates using Gauss Jordan elimination method.

$$\begin{aligned} p &\geq 55 \times \leq \\ q &\geq 15 \times \leq \\ q - p &\geq -20 \checkmark \end{aligned}$$

$$\begin{aligned} f_1 &= 55 - p \\ f_2 &= 20 + q - p \\ f_3 &= 15 - q \\ f_4 &= p \\ f_5 &= q \end{aligned}$$

Q1

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -4 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}$$

(a) $x_1 + 3x_2 = 2$

$$-2x_1 - 4x_2 + 7x_3 = 4$$

$$3x_1 + 5x_2 + 2x_3 = 1$$

$$x_1 - x_2 + 2x_3 = -3$$

(1 mark)

(b) $\begin{bmatrix} 1 & 3 & 0 & 2 \\ -2 & -4 & 7 & 4 \\ 3 & 5 & 2 & 1 \\ 1 & -1 & 2 & -3 \end{bmatrix}$

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 2 & 7 & 8 & 8 \\ -4 & 2 & -5 & -5 \\ 0 & -4 & 2 & -5 \end{bmatrix} \begin{array}{l} R_2 \leftarrow R_2 + 2R_1 \\ R_3 \leftarrow R_3 - 3R_1 \\ R_4 \leftarrow R_4 - R_1 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7/2 & 4 \\ 0 & -4 & 2 & -5 \\ 0 & -4 & 2 & -5 \end{bmatrix} R_2 \leftarrow \frac{1}{2} R_2$$

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7/2 & 4 \\ 0 & 0 & 16 & 11 \\ 0 & 0 & 16 & 11 \end{bmatrix} \begin{array}{l} R_3 \leftarrow R_3 + 4R_2 \\ R_4 \leftarrow R_4 + 4R_2 \end{array}$$

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7/2 & 4 \\ 0 & 0 & 1 & 11/16 \\ 0 & 0 & 16 & 11 \end{bmatrix} R_3 \leftarrow \frac{1}{16} R_3$$

$$\begin{bmatrix} 1 & 3 & 0 & 2 \\ 0 & 1 & 7/2 & 4 \\ 0 & 0 & 1 & 11/16 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_4 \leftarrow R_4 - 16R_3$$

$R_3 \leftarrow \frac{1}{16} R_3$

$R_4 \leftarrow R_4 - 16R_3$

$x_1 + 3x_2 = 2$ (5 marks)

$$x_2 + 7/2 x_3 = 4$$

$$x_3 = 11/16$$

$$x_2 + 7/2 \cdot 11/16 = 4$$

$$x_2 + \frac{77}{32} = 4$$

$$x_2 = 4 - \frac{77}{32}$$

$$x_2 = \frac{128 - 77}{32} = \frac{51}{32}$$

$$x_1 + 3\left(\frac{51}{32}\right) = 2$$

$$x_1 = 2 - \frac{153}{32}$$

$$x_1 = \frac{64 - 153}{32}$$

$$x_1 = -\frac{89}{32}$$

$$(x_1, x_2, x_3) = \left(-\frac{89}{32}, \frac{51}{32}, \frac{11}{16}\right)$$

(1 mark)

Q2

$$\begin{bmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{bmatrix}$$

(a) Minor $M_{21} :- M_{21} = \begin{vmatrix} 1 & 2 \\ 7 & 2 \end{vmatrix} = 2 - 14 = -12$ (2 marks)

Cofactor $C_{21} :- C_{21} = (-1)^{2+1} M_{21} = (-1)^3 (-12) = -12$ (3 marks)

(b)

$$\begin{vmatrix} -1 & 1 & 2 \\ 3 & 0 & -5 \\ 1 & 7 & 2 \end{vmatrix} = (-1)(0 + 35) - 1(6 + 5) + 2(21 - 0)$$

$$= -35 - 11 + 42$$

$$= -46 + 42$$

$$= -4$$
 (5 marks)

Q3 formula for

(a) The eq. of plane is :-

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0 \quad \rightarrow (1 \text{ mark})$$

$$1(x-1) + 9(y-1) + 8(z-4) = 0$$

$$x - 1 + 9y - 9 + 8z - 32 = 0$$

$$x + 9y + 8z - 1 - 9 - 32 = 0$$

$$x + 9y + 8z - 42 = 0$$

$$\boxed{x + 9y + 8z = 42}$$

$\rightarrow$ (4 marks)

(b) Distance formula is :-

$$d = \frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}} \quad \rightarrow (1 \text{ mark})$$

$$= \frac{|1(-1) + 9(-1) + 8(2) - 42|}{\sqrt{1^2 + 9^2 + 8^2}}$$

$$= \frac{|-1 - 9 + 16 - 42|}{\sqrt{1 + 81 + 64}}$$

$$= \frac{|-36|}{\sqrt{146}} = \frac{36}{\sqrt{146}}$$

$$= \frac{36}{\sqrt{146}}$$

$\rightarrow$ (4 marks)

Q4

$$\text{Proj}_{\vec{a}} \vec{u} = \left(\frac{\vec{u} \cdot \vec{a}}{|\vec{a}|^2} \right) \vec{a} \rightarrow (1 \text{ mark})$$

$$= \left(\frac{(2, 1, 1, 2) \cdot (4, -4, 2, -2)}{(\sqrt{4^2 + 4^2 + 2^2 + 2^2})^2} \right) (4, -4, 2, -2)$$

$$= \left(\frac{8 - 4 + 2 - 4}{(\sqrt{40})^2} \right) (4, -4, 2, -2)$$

$$= \left(\frac{\frac{2}{40}}{20} \right) (4, -4, 2, -2)$$

$$= \left(\frac{4}{20}, \frac{-4}{20}, \frac{2}{20}, \frac{-2}{20} \right)$$

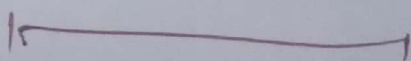
$$= \left(\frac{1}{5}, -\frac{1}{5}, \frac{1}{10}, -\frac{1}{10} \right) \rightarrow (4 \text{ mark})$$

$$\vec{u} - \text{Proj}_{\vec{a}} \vec{u} = (2, 1, 1, 2) - \left(\frac{1}{5}, -\frac{1}{5}, \frac{1}{10}, -\frac{1}{10} \right) (1 \text{ mark})$$

$$= (2 - \frac{1}{5}, 1 + \frac{1}{5}, 1 - \frac{1}{10}, 2 + \frac{1}{10})$$

$$= \left(\frac{9}{5}, \frac{6}{5}, \frac{9}{10}, \frac{21}{10} \right)$$

$$\rightarrow (4 \text{ marks})$$



National University of Computer and Emerging Sciences, Lahore Campus			
	Course:	Linear Algebra	Course Code: MT-1004
	Program:	BS(CS)/BS(DS)/BS(SE)	Semester: Fall 2022
	Duration:	60 mins	Total Marks: 40
	Date:	26-09-2022	Weight: 12.5%
	Section:	All	Page(s): 2
	Exam:	Mid 1	Roll No: _____
Name: _____			

Instruction/Notes:

Attempt all questions. If you believe that some essential piece of information is missing, make an appropriate assumption and use it to solve the problem. Use of programmable calculators is not allowed. Exchange of stationary is strictly prohibited. Best of luck!

Question no. 1: (5+5+5+3+2 marks) (CLO #1)

- a) Suppose that the augmented matrix of a system of linear equations has been partially reduced using elementary row operations to

$$\left[\begin{array}{ccc|c} 1 & -1 & a-2 & 3 \\ 0 & 1 & 2b+1 & -1 \\ 0 & 0 & a-1 & b+3 \end{array} \right]$$

-4

Find all values of a and b for which the given system has no solution, exactly one solution and infinitely many solutions.

- b) Find the matrix B by using the given information

$$(5B^T)^{-1} = \begin{bmatrix} 1 & 5 \\ 3 & -4 \end{bmatrix}$$

- c) Evaluate the given determinant by using row reduction method.

$$\begin{vmatrix} 1 & -1 & 5 & 5 \\ 3 & 1 & 2 & 4 \\ -1 & -3 & 8 & 0 \\ 1 & 1 & 2 & -1 \end{vmatrix}$$

- d) For the following linear system of equation, find the general solution in parametric form and also give the geometric representation.

$$\begin{aligned} x + 3y &= 3 \\ 2x + 6y &= 6 \end{aligned}$$

-1

- e) Identify the row operation corresponding to E and verify that the product EC results from applying the row operation to C .

$$E = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$$

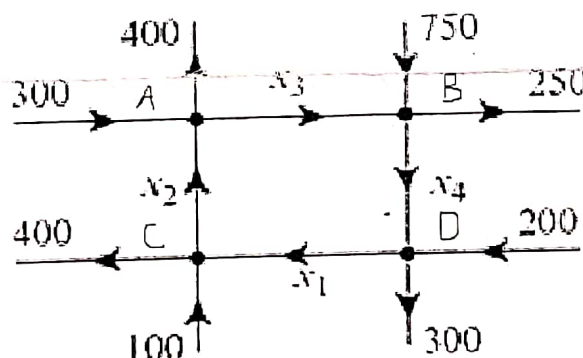
Question no. 2: (10 marks) (CLO #1)

Use inversion algorithm to find the inverse of the following matrix.

$$\begin{bmatrix} 1 & 4 & 0 \\ -2 & 2 & 3 \\ 3 & 8 & 7 \end{bmatrix}$$

Question no. 3: (10 marks) (CLO #1)

The accompanying figure shows a network of one-way streets with traffic flowing in the directions indicated. The flow rates along the streets are measured as the average number of vehicles per hour



Set up a linear system whose solution provides the unknown flow rates and solve the system for the unknown flow rates using Gauss Elimination.

National University of Computer and Emerging Sciences, Lahore Campus



Course: Linear Algebra
Program: BS (CS, DS, SE)
Duration: 60 Minutes
Paper Date: 19-Oct-21
Section: ALL
Exam: Midterm-I

Course Code: MT1004
Semester: Fall 2021
Total Marks: 30
Weight: 12.5%
Page(s): 1
Roll No:

Instruction/Notes: Programmable calculators are not allowed.

Question # 1(a) [5]: Suppose three households in your neighborhood performed holy sacrifice this Eid. The 1st household sacrificed 2 cows, 4 lambs and 1 camel. The 2nd household sacrificed 1 cow, 3 lambs and 1 camel. The 3rd household sacrificed 3 cows, 5 lambs and 1 camel. If the first household spent total Rs. 9, 00, 000, second household spent Rs. 6,50,000 and third household spent Rs. 11,50,000. Use **Gauss Jordan Elimination** to determine how much each cow, lamb and camel cost?

Question # 1(b) [5]: Suppose that the **Augmented matrix** of a linear system of equations has been reduced by row operations to the given **Row Echelon** form. By inspection find the solution of the system from the given form.

$$\begin{bmatrix} 1 & 7 & -2 & 0 & -8 & -3 \\ 0 & 0 & 1 & 1 & 6 & 5 \\ 0 & 0 & 0 & 1 & 3 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Question # 2 [10]: Determine condition(s) on b_i 's, if any, in order to guarantee that the linear system is consistent

$$\begin{aligned} x_1 - 2x_2 - x_3 &= b_1 \\ 2x_1 + 5x_2 + 2x_3 &= b_2 \\ -3x_1 + 7x_2 + 4x_3 &= b_3 \end{aligned}$$

Question # 3 [6+4]: Use Inverse Algorithm (By Elementary Row Operations) to find the inverse of the matrix A. Also, express B as a product of the elementary matrices.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

National University of Computer and Emerging Sciences, Lahore Campus



Course: Linear Algebra
Program: BS(CS)
Duration: 60 Minutes
Paper Date: September-19
Section: ALL
Exam: Midterm-I

Course Code: MT104
Semester: Fall 2019
Total Marks: 30
Weight: 12.5
Page(s): 1
Roll No: [REDACTED]

Instruction/Notes: Attempt All Questions.

Question # 1: (CLO: 1,2, 3) If $A = \begin{pmatrix} -1 & 7 & -1 \\ 0 & 1 & 0 \\ 0 & 15 & -2 \end{pmatrix}$

- a) [10] Find A^{-1} by using inversion algorithm,
- b) [10] Show that A can be expressible as a product of elementary matrices,

c) [2] If $AX = b$ and $X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$, $b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then find the value of X by using A^{-1}

d) [3] Find an elementary matrix E that satisfies the stated equation

$$E A = \begin{bmatrix} -1 & 7 & -1 \\ 0 & 1 & 0 \\ -2 & 29 & -4 \end{bmatrix}$$

e) [5] Use the row reduction to evaluate the determinant of $B = \begin{bmatrix} 1 & 3 & 1 & 5 & 3 \\ -2 & -7 & 0 & -4 & 2 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$